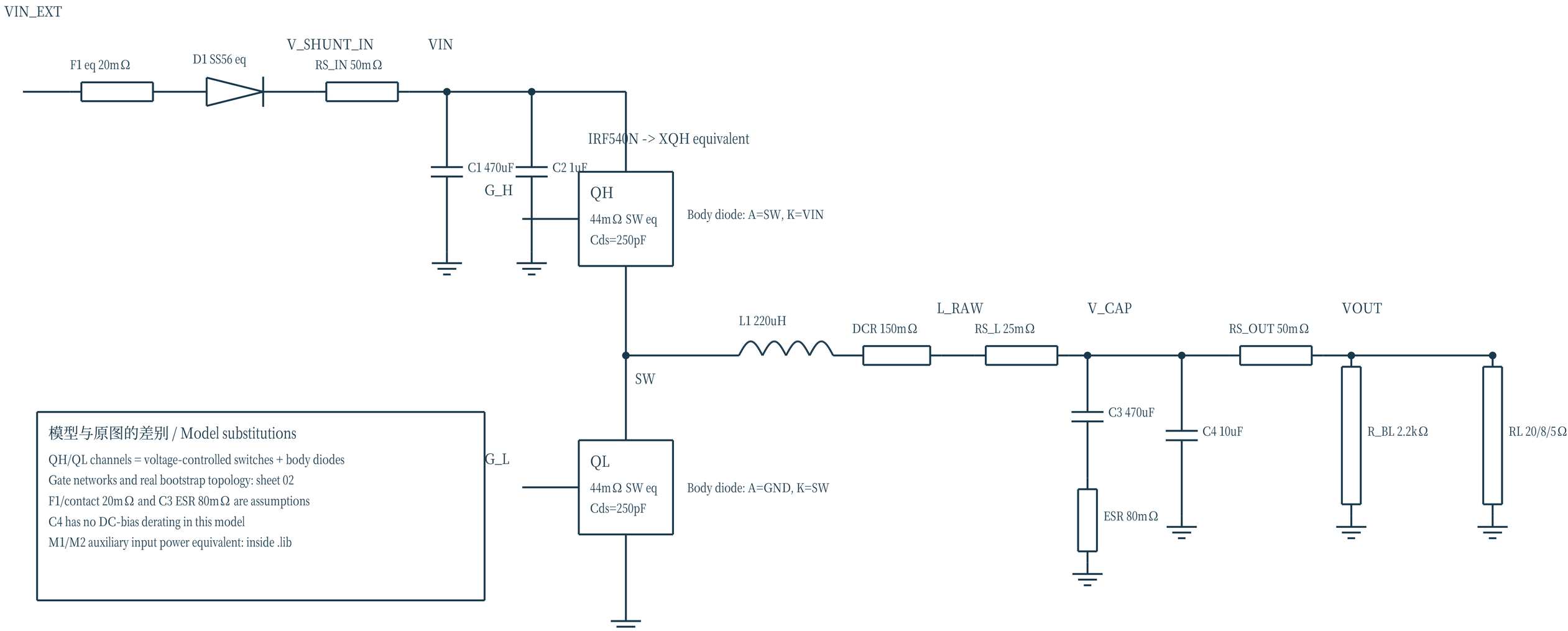
01 主功率接线 / Main power stage

Rev B PDF reconstruction | Functional equivalents | Not a manufacturer model or native Multisim schematic



模型与原图的差别 / Model substitutions

- QH/QL channels = voltage-controlled switches + body diodes
- Gate networks and real bootstrap topology: sheet 02
- F1/contact 20mΩ and C3 ESR 80mΩ are assumptions
- C4 has no DC-bias derating in this model
- M1/M2 auxiliary input power equivalent: inside .lib

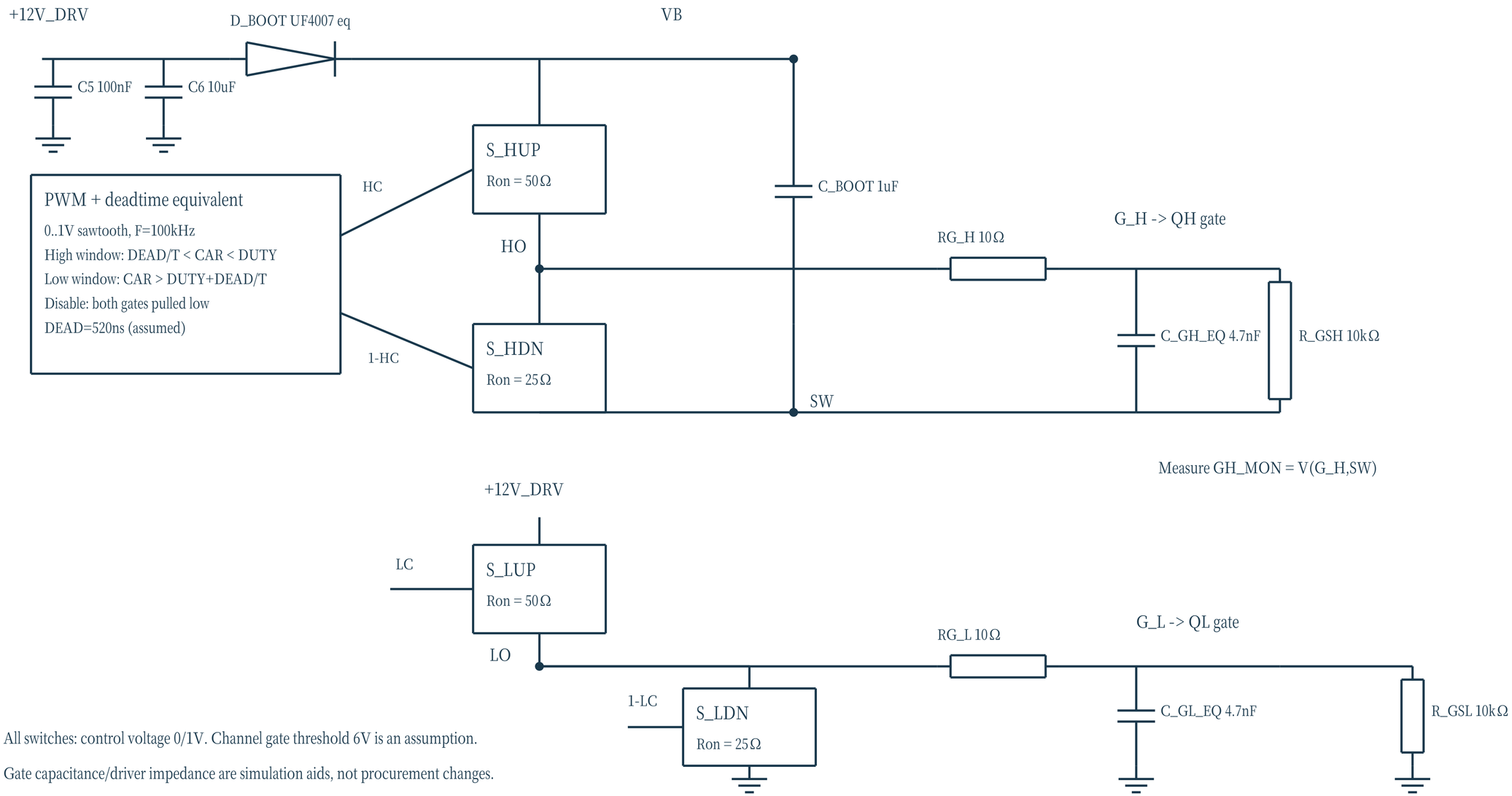
电流采样极性：U2 IN+ = V_SHUNT_IN, IN- = VIN; U3 IN+ = V_CAP, IN- = VOUT

U4 IN+ = L_RAW, IN- = V_CAP. 25mΩ × gain 20 = 0.5V/A; zero = V3A/2.

02 浮动高侧驱动、自举与互锁 / Floating driver and bootstrap

Rev B PDF reconstruction | Functional equivalents | Not a manufacturer model or native Multisim schematic

这是IR2104功能替代电路，不是厂家IR2104模型。高侧必须相对SW驱动，禁止把高侧栅极只接对地12V。



All switches: control voltage 0/1V. Channel gate threshold 6V is an assumption.

Gate capacitance/driver impedance are simulation aids, not procurement changes.

Generated 2026-10-22 | Pin order and model assumptions: ../README.md | COM must be connected to SPICE ground 0

03 采样、闭环与保护 / Sensing, continuous control and protection

Rev B PDF reconstruction | Functional equivalents | Not a manufacturer model or native Multisim schematic

电压环 / CV

VSET × soft-start - 11×ADC_VOUT
PI: Kp=0.5, Ki=50
Anti-windup: 500/s

电流目标限幅

IREF = LIMIT(PI+FF,-0.3,ISET)
ISET=1V represents 1A
FF=Cout × dVset/dt

电感电流环 / IL

ILmeas=2×(ADC_IL - V3A/2)
PI: Kp=0.15, Ki=200
Feedforward: V_CAP/VIN

PWM

DUTY = LIMIT(raw,0.08,0.88)
Compare with 100kHz CAR
Deadtime + enable -> driver

功率级 / Plant

See sheet 01 and 02
VOUT + inductor current
Sources of feedback

电流测量 / Current measurement

U2: IIN_RAW = 20 × V(RS_IN) = 1V/A
U3: IOUT_RAW = 20 × V(RS_OUT) = 1V/A
U4: IL_BI = V3A/2 + 20 × V(RS_L) = V3A/2+0.5V/A
Each ADC branch: 1kΩ + 4.7nF to GND
Amplifier output 0..V3A; raw pole ~0.4us (assumed)

电压采样 / Voltage measurement

VIN_EXT --100k-- VIN_DIV --10k-- GND
VIN_DIV --1k-- ADC_VIN; 4.7nF to GND
VOUT --10k-- VOUT_DIV --1k-- GND
VOUT_DIV --1k-- ADC_VOUT; 4.7nF to GND
V3 ->10Ω->V3A; 1uF+100nF+10uF to GND

参考网络 / References (all 100nF to GND)

V3A --10k-- VREF_POS --33k--1k-- GND
V3A --33k--1k-- VREF_NEG --10k-- GND
V3A --23k-- VREF_OV --10k-- GND
At V3A=3.3V: POS=2.55V, NEG=0.75V, OV=1.00V
Actual V3A is lower because R_VDDA carries current

比较器功能等效 / Fault comparators

IL_BI > VREF_POS -> FAULT_N low
IL_BI < VREF_NEG -> FAULT_N low
VOUT_DIV > VREF_OV -> FAULT_N low
Outputs wired-OR, 10k pull-up to 3.3V
1.3us output RC = assumption, not guaranteed delay

锁存与使能 / Latch and enable

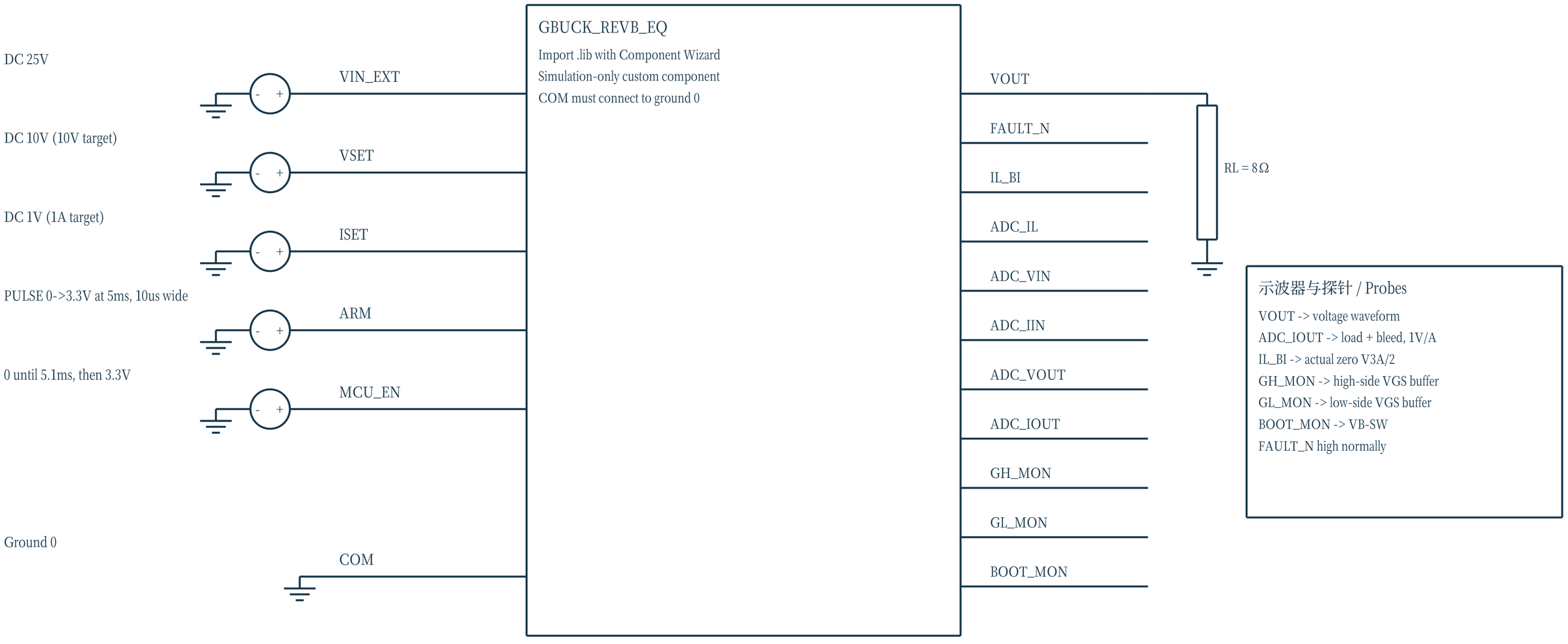
POR holds reset until 4ms (functional approximation)
FAULT_N low asynchronously clears SAFE
ARM rising edge sets SAFE after fault clears
DRV_EN = SAFE AND MCU_EN
FAULT cleared alone does not restart switching

重要边界 / Not validated

Continuous PI is not F103 firmware
No LCD/key/ADC sampling-time simulation
No actual HC74/HC14 setup/hold/ramp validation
No vendor transistor/Miller/saturation model
No native Multisim 14 execution on this host

04 Multisim首个可搭建电路 / 17-pin component testbench

Rev B PDF reconstruction | Functional equivalents | Not a manufacturer model or native Multisim schematic



Transient: 0..80ms, maximum step 100ns. Choose zero initial capacitor voltage/inductor current (UIC).

Expected: about 8V / 1A. The input 25V is supplied only to VIN_EXT; no separate external 12V/5V rails are needed.